

Coupling Interpretable Allocation, Model-Predictive Control, and Tail-Aware Forecasting for Live Portfolio Management

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Utility functions turn a preference score into weights

Stock i: n_i shares at price S_i, weight $w_i = n_i S_i / W$ in a book of wealth W. A single-index model of daily growth rates gives α_i and β_i. With market growth signal $\tilde{g}_M$, market-state signal ξ from SPY's moving averages and $\beta_{\text{floor}} = 0.1$, the preference weight is

$$\gamma_i = \tanh\big((\alpha_i + \beta_i \tilde{g}_M) \,/\, \max(|\beta_i|, \beta_{\text{floor}})^{\xi}\big).$$

Choose n to maximize $U(n;\gamma)$ subject to:

- the budget;
- at most 10% of wealth per name;
- volatility capped at SPY's 2014–2024 level;
- a turnover budget per rebalance;
- hedge on: a cap on the portfolio β.

Utilities, summed over $\gamma_i > 0$, with $\eta > 0$ the CES elasticity, $\rho = 1 - 1/\eta$ and b_i a minimum holding:

- Cobb-Douglas $\sum \gamma_i \ln n_i$;
- CES $\big(\sum \gamma_i n_i^{\rho}\big)^{1/\rho}$;
- CES on weights, the same with w_i;
- Stone–Geary $\sum \gamma_i \ln(n_i - b_i)$.

Each solve is a convex cone program: exponential cones for the log utilities, power cones for CES. Also compared: minimum variance, the tangency portfolio, and 130/30 long-short Cobb-Douglas.

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Under the budget alone, with $\eta > 0$ the CES elasticity, the closed forms:

- Cobb-Douglas $w_i \propto \gamma_i$;
- CES on weights $w_i \propto \gamma_i^{\eta}$;
- CES on shares $n_i \propto (\gamma_i / S_i)^{\eta}$.

No solver needed, so allocation is instant.

Also compared: minimum variance, the tangency portfolio, and 130/30 long-short Cobb-Douglas.

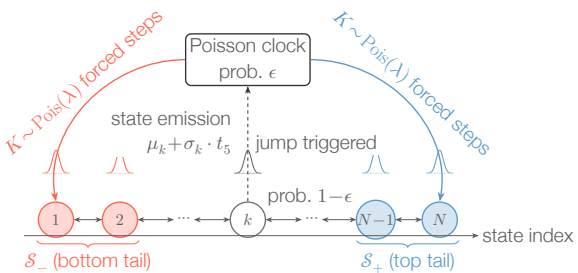
When to rebalance: re-plan when the market leaves the forecast

Controller's model: multi-asset GBM, 1,000 paths, 21 trading days. Plant: real prices.
Re-plan when the value leaves the 95% band, falls 8% below its peak, or 21 trading days pass.

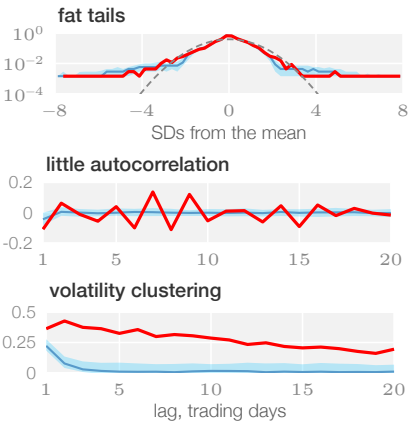


Real-data backtest: hand-picked 13 names, Cobb-Douglas, hedge off. Bands are the forecasts stored at each rebalance. Over the full backtest: about 27 rebalances after the first allocation in 419 trading days.

JumpHMM: a market generator with fat tails



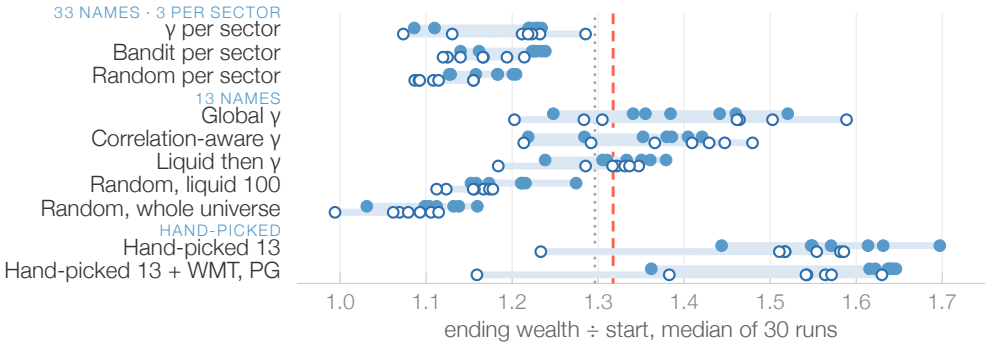
Alswaidan and Varner, arXiv:2603.10202



Real SPY daily growth, 2014–2024 (red), against 100 paths of this year's generator (blue, 10th to 90th percentile). In the generator, volatility clustering fades after a day; in real SPY it lasts weeks.

What you hold beats how you weight it

Real-data backtest, Jan 2025 – Sep 2026

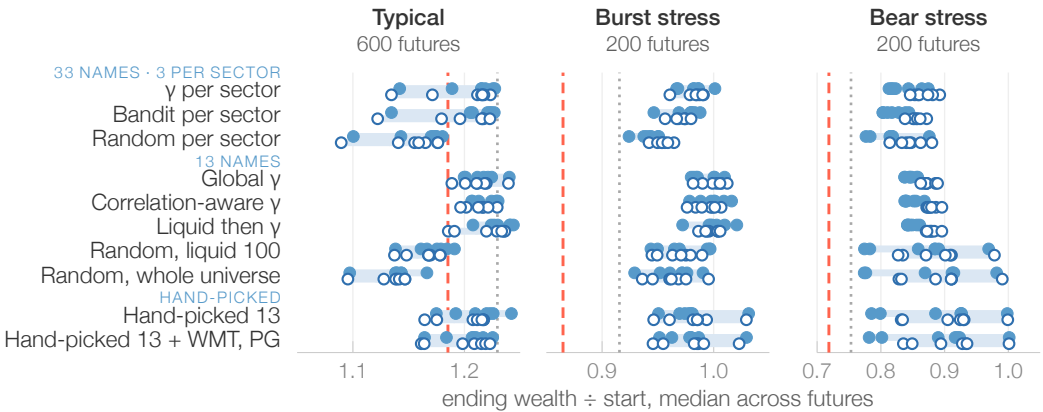


● hedge off ○ hedge on — · — SPY buy-and-hold ····· equal-weight universe one dot per utility

hedge on: up to 5 names with negative β join the basket and the book's β is capped

A paper-trading deployment of this system is running; it is not part of this talk.

978 synthetic futures in three regimes



Key takeaways

- **Utility functions set the weights.** We compared seven weighting rules: four utility functions (Cobb-Douglas, CES on shares, CES on weights, Stone–Geary), minimum variance, the tangency portfolio, and a 130/30 long-short Cobb-Douglas. Under a budget alone, the utility rules have a closed-form answer. With limits on single names, sectors, volatility and turnover, the problem stays convex and a standard solver handles it. In the real run, the stocks we picked mattered more than the rule: on one 33-stock basket the seven rules ended between 1.09 and 1.23 times the starting wealth, while different baskets ranged from 1.03 to 1.70.
- **The MPC trigger decides when to rebalance.** After each rebalance we simulate 1,000 paths of the portfolio's value over the next 21 trading days. We rebalance early if the value leaves the 95% band or falls 8% below its peak, and otherwise wait out the 21 days. In the real run that came to about 27 rebalances in 419 trading days.
- **We tested the method on simulated markets.** A JumpHMM fitted to 2014–2024 data reproduces fat tails, weak autocorrelation and short-lived volatility clustering. We used it to simulate 978 markets: 600 typical, 200 with jump bursts and 200 deep bear markets. Picking 13 stocks by γ beat 13 random stocks in 62–71% of typical markets. In bear markets, random picks did as well or better for most weighting rules.

Acknowledgements

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- **Aaron Wheeler**, PhD 2024
  helpful discussions
- **Evan Vu**, undergraduate, class of 2026, now at Bridgewater
  an early version of the Cobb-Douglas allocation and the bandit picker

Alpaca Markets

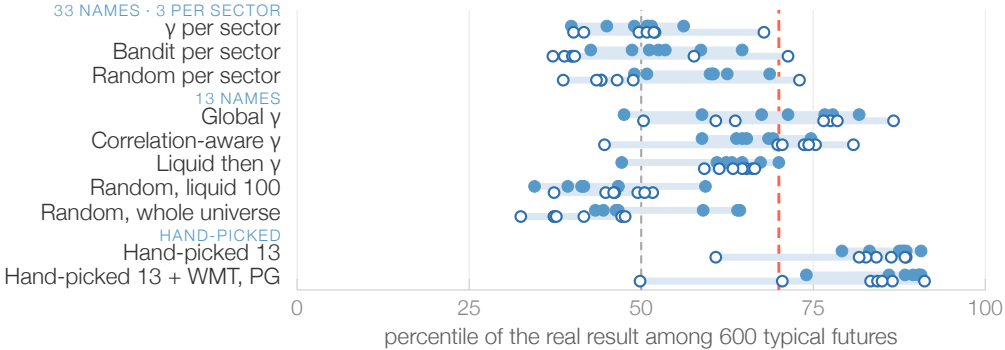
Paper

JumpHMM paper
arXiv:2603.10202

Code

github.com/varnerlab/INFORMS-Annual-Meeting-2026-Varnerlab

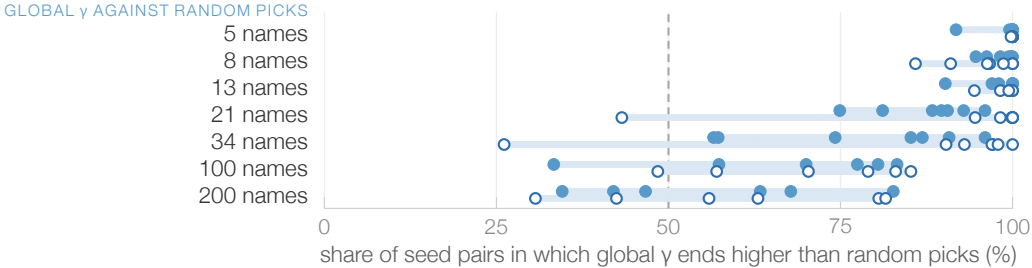
Was 2025 a lucky year?



hedge off hedge on SPY buy-and-hold one dot per utility

Real results use the same seeds as the simulated runs: 5660–5664 for random and bandit pickers, 5660 for the others.

Does this work for a real-sized book?



Real-data backtest, 30 runs each. Stone–Geary left out at 100 and 200 names: its 1%-per-name floor uses the whole budget at 100 names and twice the budget at 200.

Show me the numbers: hedge off

Picker	CD	CES	CESw	SG	MinVar	Tangent	LS
γ per sector	1.230	1.219	1.221	1.234	1.086	1.227	1.110
Bandit per sector	1.226	1.234	1.223	1.239	1.162	1.229	1.140
Random per sector	1.205	1.183	1.201	1.183	1.128	1.158	1.127
Global γ	1.520	1.384	1.460	1.441	1.355	1.341	1.248
Correlation-aware γ	1.352	1.386	1.421	1.380	1.284	1.404	1.219
Liquid then γ	1.333	1.310	1.379	1.350	1.238	1.361	1.305
Random, liquid 100	1.216	1.173	1.211	1.212	1.158	1.274	1.152
Random, whole universe	1.098	1.132	1.138	1.112	1.104	1.160	1.031
Hand-picked 13	1.548	1.571	1.614	1.631	1.443	1.697	1.549
Hand-picked 13 + WMT, PG	1.616	1.646	1.637	1.639	1.362	1.642	1.622
SPY buy-and-hold	1.317	1.317	1.317	1.317	1.317	1.317	1.317
equal-weight universe	1.296	1.296	1.296	1.296	1.296	1.296	1.296

Ending wealth ÷ start, median of 30 runs. Real-data backtest, Jan 2025 – Sep 2026.

Show me the numbers: hedge on

Picker	CD	CES	CESw	SG	MinVar	Tangent	LS
γ per sector	1.212	1.232	1.223	1.218	1.073	1.285	1.130
Bandit per sector	1.125	1.140	1.167	1.194	1.166	1.214	1.120
Random per sector	1.108	1.087	1.114	1.155	1.155	1.092	1.092
Global γ	1.503	1.305	1.464	1.461	1.203	1.589	1.283
Correlation-aware γ	1.409	1.429	1.479	1.447	1.213	1.366	1.292
Liquid then γ	1.322	1.285	1.347	1.331	1.184	1.317	1.336
Random, liquid 100	1.167	1.156	1.174	1.178	1.123	1.155	1.112
Random, whole universe	1.069	1.106	1.093	1.115	1.079	1.062	0.994
Hand-picked 13	1.554	1.517	1.517	1.511	1.233	1.582	1.586
Hand-picked 13 + WMT, PG	1.564	1.571	1.543	1.542	1.159	1.383	1.630
SPY buy-and-hold	1.317	1.317	1.317	1.317	1.317	1.317	1.317
equal-weight universe	1.296	1.296	1.296	1.296	1.296	1.296	1.296

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